DRUGS · ANTIBIOTICS: PENICILLINS

# Piperacillin with tazobactam

Piperacillin with tazobactam is a broad-spectrum ureidopenicillin protected by a beta-lactamase inhibitor. It covers Pseudomonas, most Enterobacterales, staphylococci (not MRSA), streptococci and anaerobes, which makes it a standard intravenous choice for necrotising otitis externa, hospital-acquired head and neck infection and sepsis after major surgery. It is a hospital drug, and every use should be reviewed at 48 to 72 hours.

## 1 CHAPTER ONE At a glance

The drug on one page.

### 1.1 Summary

PIPERACILLIN WITH TAZOBACTAM

| | SUMMARY |
|---|---|
| **Class** | Ureidopenicillin with a beta-lactamase inhibitor |
| **How it works** | Cell wall synthesis inhibition; tazobactam protects against many beta-lactamases |
| **Spectrum** | Pseudomonas aeruginosa, Enterobacterales, methicillin-sensitive S. aureus, streptococci, enterococci (some), anaerobes; not MRSA, not ESBL reliably, not atypicals |
| **ENT uses** | Necrotising (malignant) otitis externa and skull base osteomyelitis; hospital-acquired and post-operative deep neck and wound infection; aspiration pneumonia after head and neck surgery; febrile neutropenia |
| **Adult dose** | 4.5 g every 8 hours intravenously, increased to every 6 hours in severe infection; extended infusions over 3 to 4 hours improve time above the MIC |
| **Child dose** | 90 mg/kg every 6 to 8 hours (maximum 4.5 g per dose) |
| **Duration** | Review at 48 to 72 hours with cultures; necrotising otitis externa needs 6 weeks or more of anti-pseudomonal therapy, usually switched to oral ciprofloxacin |

| | SUMMARY |
|---|---|
| **Avoid** | Penicillin allergy |
| **Pregnancy** | Use if needed |

# 2 CHAPTER TWO How it works

Mechanism and handling.

## 2.1 Mechanism

Piperacillin's side chain lets it penetrate the outer membrane of Gram-negative bacilli including Pseudomonas; tazobactam inhibits many plasmid-mediated beta-lactamases. Extended-spectrum beta-lactamase (ESBL) and AmpC producers are often resistant; use a carbapenem for proven ESBL infection.

## 2.2 Handling

Given intravenously only; renally excreted, so reduce the dose when the creatinine clearance is below 40 ml/min.

# 3 CHAPTER THREE Using it in ENT

Indications and doses.

## 3.1 Necrotising otitis externa

An elderly diabetic or immunocompromised patient with severe otalgia out of proportion, granulation at the bony-cartilaginous junction and cranial nerve palsy (VII first). Take deep cultures and biopsy (to exclude carcinoma) **before** antibiotics if possible, control the diabetes, start intravenous anti-pseudomonal therapy (piperacillin-tazobactam or ceftazidime, often with ciprofloxacin), and convert to

oral ciprofloxacin for a total of at least 6 weeks guided by clinical response, inflammatory markers and imaging.

### 3.2 Post-operative and hospital infection

After major head and neck surgery, a wound or neck infection with a pharyngocutaneous fistula, or aspiration pneumonia, is often polymicrobial with Gram-negatives; piperacillin-tazobactam is a common empirical choice, narrowed when cultures return.

## 4 CHAPTER FOUR Safety

What goes wrong.

### 4.1 Adverse effects

- Hypersensitivity; diarrhoea and C. difficile.
- **Acute kidney injury**, especially in combination with vancomycin.
- Hypokalaemia (sodium load), thrombocytopenia and neutropenia with prolonged courses, and hepatitis.

### 4.2 Contraindications and cautions

- Penicillin allergy.
- Renal impairment (dose reduction).

### 4.3 Interactions

**Vancomycin** (additive nephrotoxicity: monitor renal function); methotrexate; aminoglycosides (inactivated if mixed in the same infusion: give separately).

# 5 CHAPTER FIVE For the examination

What the examiner wants to hear.

## 5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Organisms of **necrotising otitis externa** (Pseudomonas in most; Aspergillus in some) and the treatment plan.
- Which beta-lactams cover Pseudomonas: piperacillin-tazobactam, ceftazidime, cefepime, carbapenems (not ertapenem).
- Why to culture and biopsy before treatment.

## 5.2 Common pitfalls

COMMON PITFALLS

- Starting antibiotics in necrotising otitis externa before taking deep cultures and a biopsy.
- Stopping too early on clinical improvement.
- Continuing broad-spectrum therapy without a 48-hour review.

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